

19-2 IGP Fundamentals Troubleshooting – Answer Key

In this lab you will troubleshoot the RIPv2 and EIGRP routing protocols.

RIP Troubleshooting

- 1) RIPv2 has been configured on the network but the routers are not learning remote routes.

Your task is to troubleshoot the configuration and ensure that all routers have routes to every 10.x.x.x network.

There is a configuration error on each router. You will need to find and fix all errors before the network will work properly.

There is not a set order of actions to troubleshoot this scenario. Troubleshooting in a logical fashion will however make it easier and quicker. This is how I would do it.

Check to verify that there are no RIP routes on any router. Always verify the reported problem exists first, this can save you a lot of wasted time spent troubleshooting non-existent issues.

R1#sh ip route

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * - candidate default, U - per-user static route
o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
+ - replicated route, % - next hop override

Gateway of last resort is not set

```
10.0.0.0/8 is variably subnetted, 8 subnets, 2 masks
C      10.0.0.0/24 is directly connected, FastEthernet0/0
L      10.0.0.1/32 is directly connected, FastEthernet0/0
C      10.0.1.0/24 is directly connected, FastEthernet0/1
L      10.0.1.1/32 is directly connected, FastEthernet0/1
C      10.0.2.0/24 is directly connected, FastEthernet1/0
L      10.0.2.1/32 is directly connected, FastEthernet1/0
C      10.0.3.0/24 is directly connected, FastEthernet1/1
L      10.0.3.1/32 is directly connected, FastEthernet1/1
```

None of the routers have any RIP routes. R5 is different in that it only has a route for its loopback interface, there are no connected routes, so we'll start here.

```
R5#sh ip route
```

```
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2
        i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
        ia - IS-IS inter area, * - candidate default, U - per-user static route
        o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
        + - replicated route, % - next hop override
```

Gateway of last resort is not set

192.168.0.0/32 is subnetted, 1 subnets

C 192.168.0.5/32 is directly connected, Loopback0

No connected routes indicates there might be a problem with the interfaces. Check this.

```
R5#show ip interface brief
```

Interface	IP-Address	OK?	Method	Status
Protocol				
FastEthernet0/0	10.1.3.2	YES	NVRAM	administratively down down
FastEthernet0/1	10.0.3.2	YES	NVRAM	administratively down down
FastEthernet1/0	unassigned	YES	NVRAM	administratively down down
FastEthernet1/1	unassigned	YES	NVRAM	administratively down down
Loopback0	10.1.3.2	YES	manual	up up

The FastEthernet 0/0 and 0/1 interfaces are shut down. Bring them back up again.

```
R5(config)#int f0/0
```

```
R5(config-if)#no shutdown
```

```
R5(config)#int f0/1
```

```
R5(config-if)#no shutdown
```

Check the routing table now.

```
R5#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route
```

Gateway of last resort is not set

```

    10.0.0.0/8 is variably subnetted, 4 subnets, 2 masks
C       10.0.3.0/24 is directly connected, FastEthernet0/1
L       10.0.3.2/32 is directly connected, FastEthernet0/1
C       10.1.3.0/24 is directly connected, FastEthernet0/0
L       10.1.3.2/32 is directly connected, FastEthernet0/0
    192.168.0.0/32 is subnetted, 1 subnets
C       192.168.0.5/32 is directly connected, Loopback0
```

We have connected routes now but no RIP routes. Check the RIP configuration.

```
R5#sh run | section rip
router rip
version 2
network 10.0.0.0
no auto-summary
```

That looks okay. Let's dig deeper.

```
R5#show ip protocols
! truncated (showing only RIP)
*** IP Routing is NSF aware ***
Routing Protocol is "rip"
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Sending updates every 30 seconds, next due in 18 seconds
  Invalid after 180 seconds, hold down 180, flushed after 240
  Redistributing: rip
  Default version control: send version 2, receive version 2
    Interface          Send  Recv  Triggered RIP  Key-chain
    FastEthernet2/0      2      2
    FastEthernet3/0      2      2
  Automatic network summarization is not in effect
  Maximum path: 4
  Routing for Networks:
    10.0.0.0
  Routing Information Sources:
    Gateway          Distance          Last Update
  Distance: (default is 120)
```

That looks okay too. Let's move on to R5's neighbour R1 and check its RIP configuration.

```
R1#sh run | sec rip
router rip
version 2
network 100.0.0.0
no auto-summary
```

Hopefully you can see the error. If not, this might help:

```
R1#show ip protocols
! truncated (showing only RIP)
*** IP Routing is NSF aware ***
```

```
Routing Protocol is "rip"
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Sending updates every 30 seconds, next due in 0 seconds
  Invalid after 180 seconds, hold down 180, flushed after 240
  Redistributing: rip
  Default version control: send version 2, receive version 2
  Automatic network summarization is not in effect
  Maximum path: 4
Routing for Networks:
  100.0.0.0
  Routing Information Sources:
    Gateway         Distance         Last Update
  Distance: (default is 120)
```

RIP is not enabled on any interfaces. This is because a typo was entered – 'network 100.0.0.0' instead of 'network 10.0.0.0'. Fix this.

```
R1(config)#router rip
R1(config-router)#no network 100.0.0.0
R1(config-router)#network 10.0.0.0
```

'Show ip protocols' should now show RIP enabled on the interfaces.

```
R1#show ip protocols
*** IP Routing is NSF aware ***
```

```
Routing Protocol is "rip"
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Sending updates every 30 seconds, next due in 25 seconds
  Invalid after 180 seconds, hold down 180, flushed after 240
  Redistributing: rip
  Default version control: send version 2, receive version 2
    Interface          Send  Recv  Triggered RIP  Key-chain
  FastEthernet0/0      2     2
  FastEthernet0/1      2     2
  FastEthernet1/0      2     2
  FastEthernet1/1      2     2
  Automatic network summarization is not in effect
  Maximum path: 4
  Routing for Networks:
    10.0.0.0
  Routing Information Sources:
    Gateway           Distance      Last Update
    10.0.3.2           120          00:00:00
  Distance: (default is 120)
```

R1 and R5 should now be learning RIP routes from each other.

```
R1#sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       + - replicated route, % - next hop override
```

Gateway of last resort is not set

```
10.0.0.0/8 is variably subnetted, 9 subnets, 2 masks
C    10.0.0.0/24 is directly connected, FastEthernet0/0
L    10.0.0.1/32 is directly connected, FastEthernet0/0
C    10.0.1.0/24 is directly connected, FastEthernet0/1
L    10.0.1.1/32 is directly connected, FastEthernet0/1
C    10.0.2.0/24 is directly connected, FastEthernet1/0
L    10.0.2.1/32 is directly connected, FastEthernet1/0
C    10.0.3.0/24 is directly connected, FastEthernet1/1
L    10.0.3.1/32 is directly connected, FastEthernet1/1
R    10.1.3.0/24 [120/1] via 10.0.3.2, 00:00:25, FastEthernet1/1
    192.168.0.0/32 is subnetted, 1 subnets
C    192.168.0.1/32 is directly connected, Loopback0
```

```
R5#sh ip route
```

```
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2
        i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
        ia - IS-IS inter area, * - candidate default, U - per-user static route
        o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
        + - replicated route, % - next hop override
```

Gateway of last resort is not set

```
      10.0.0.0/8 is variably subnetted, 7 subnets, 2 masks
R       10.0.0.0/24 [120/1] via 10.0.3.1, 00:00:10, FastEthernet0/1
R       10.0.1.0/24 [120/1] via 10.0.3.1, 00:00:10, FastEthernet0/1
R       10.0.2.0/24 [120/1] via 10.0.3.1, 00:00:10, FastEthernet0/1
C       10.0.3.0/24 is directly connected, FastEthernet0/1
L       10.0.3.2/32 is directly connected, FastEthernet0/1
C       10.1.3.0/24 is directly connected, FastEthernet1/1
L       10.1.3.2/32 is directly connected, FastEthernet1/1
      192.168.0.0/32 is subnetted, 1 subnets
C 192.168.0.1/32 is directly connected, Loopback0
```

They are still not learning RIP routes from other routers though. Check the RIP configuration on R5's neighbour R4.

```
R4#sh run | sec rip
router rip
version 2
passive-interface default
network 10.0.0.0
no auto-summary
```

All interfaces are passive by default. Check to see if this has been overridden at the interface level.

```
R4#sh run
!
interface FastEthernet0/0
ip address 10.1.1.1 255.255.255.0
duplex auto
speed auto
!
! Truncated
```

It hasn't. The only interface which should be configured as passive is FastEthernet 1/1 facing the service provider. Configure it as passive and remove the passive default command.

```
R4(config)#router rip
R4(config-router)#no passive-interface default
R4(config-router)#passive-interface f1/1
```

R5 and R3 should now receive RIP routes from R4.

```
R5#sh ip route
```

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * - candidate default, U - per-user static route
o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
+ - replicated route, % - next hop override

Gateway of last resort is not set

```
10.0.0.0/8 is variably subnetted, 9 subnets, 2 masks
R    10.0.0.0/24 [120/1] via 10.0.3.1, 00:00:21, FastEthernet0/1
R    10.0.1.0/24 [120/1] via 10.0.3.1, 00:00:21, FastEthernet0/1
R    10.0.2.0/24 [120/1] via 10.0.3.1, 00:00:21, FastEthernet0/1
C    10.0.3.0/24 is directly connected, FastEthernet0/1
L    10.0.3.2/32 is directly connected, FastEthernet0/1
R    10.1.1.0/24 [120/1] via 10.1.3.1, 00:00:27, FastEthernet0/0
R    10.1.2.0/24 [120/1] via 10.1.3.1, 00:00:27, FastEthernet0/0
C    10.1.3.0/24 is directly connected, FastEthernet0/0
L    10.1.3.2/32 is directly connected, FastEthernet0/0
192.168.0.0/32 is subnetted, 1 subnets
C 192.168.0.1/32 is directly connected, Loopback0
```

A check of R4's routing table will show it is receiving RIP routes from R5 but not from R3.

R4#sh ip route

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * - candidate default, U - per-user static route
o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
+ - replicated route, % - next hop override

Gateway of last resort is not set

```

      10.0.0.0/8 is variably subnetted, 10 subnets, 2 masks
R       10.0.0.0/24 [120/2] via 10.1.3.2, 00:00:27, FastEthernet1/0
R       10.0.1.0/24 [120/2] via 10.1.3.2, 00:00:27, FastEthernet1/0
R       10.0.2.0/24 [120/2] via 10.1.3.2, 00:00:27, FastEthernet1/0
R       10.0.3.0/24 [120/1] via 10.1.3.2, 00:00:28, FastEthernet1/0
C       10.1.1.0/24 is directly connected, FastEthernet0/0
L       10.1.1.1/32 is directly connected, FastEthernet0/0
C       10.1.2.0/24 is directly connected, FastEthernet0/1
L       10.1.2.1/32 is directly connected, FastEthernet0/1
C       10.1.3.0/24 is directly connected, FastEthernet1/0
L       10.1.3.1/32 is directly connected, FastEthernet1/0
      192.168.0.0/32 is subnetted, 1 subnets
C 192.168.0.4/32 is directly connected, Loopback0
      203.0.113.0/24 is variably subnetted, 2 subnets, 2 masks
C       203.0.113.0/24 is directly connected, FastEthernet1/1
L       203.0.113.1/32 is directly connected, FastEthernet1/1
S* 0.0.0.0/0 [1/0] via 203.0.113.2
```

Check the RIP configuration on R3.

```
R3#sh run | sec rip
router rip
version 2
network 10.0.0.0
no auto-summary
```

That looks okay.

Check the RIP database to see if routes are being received.

```
R3#show ip rip database
10.1.0.0/24 auto-summary
10.1.0.0/24 directly connected, FastEthernet0/1
10.1.1.0/24 auto-summary
10.1.1.0/24 directly connected, FastEthernet0/0
```

The RIP database only contains local routes, not routes from neighbors. We've already verified the RIP configuration on R4 so it seems there's another problem here on R3.

Check the interfaces.

```
R3#show ip interface brief
```

Interface	IP-Address	OK?	Method	Status	Protocol
FastEthernet0/0	10.1.1.1	YES	manual	up	up
FastEthernet0/1	10.1.0.2	YES	manual	up	up
FastEthernet1/0	unassigned	YES	manual	administratively down	down
FastEthernet1/1	unassigned	YES	NVRAM	administratively down	down
Loopback0	192.168.0.3	YES	manual	up	up
Vlan1	unassigned	YES	unset	administratively down	down

This one is difficult to catch but if you check closely you'll see that the IP addresses have been misconfigured on R3. They are duplicate IP addresses with its neighbour routers. Correct the IP addresses on R3.

```
R3(config)#int f0/0
R3(config-if)#ip address 10.1.1.2 255.255.255.0
R3(config)#int f0/1
R3(config-if)#ip address 10.1.0.1 255.255.255.0
```

Check that R4 is now receiving routes from R3.

R4#sh ip route

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * - candidate default, U - per-user static route
o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
+ - replicated route, % - next hop override

Gateway of last resort is not set

```

      10.0.0.0/8 is variably subnetted, 11 subnets, 2 masks
R       10.0.0.0/24 [120/2] via 10.1.3.2, 00:00:07, FastEthernet1/0
R       10.0.1.0/24 [120/2] via 10.1.3.2, 00:00:07, FastEthernet1/0
R       10.0.2.0/24 [120/2] via 10.1.3.2, 00:00:07, FastEthernet1/0
R       10.0.3.0/24 [120/1] via 10.1.3.2, 00:00:07, FastEthernet1/0
R       10.1.0.0/24 [120/1] via 10.1.1.2, 00:00:22, FastEthernet0/0
C       10.1.1.0/24 is directly connected, FastEthernet0/0
L       10.1.1.1/32 is directly connected, FastEthernet0/0
C       10.1.2.0/24 is directly connected, FastEthernet0/1
L       10.1.2.1/32 is directly connected, FastEthernet0/1
C       10.1.3.0/24 is directly connected, FastEthernet1/0
L       10.1.3.1/32 is directly connected, FastEthernet1/0
      192.168.0.0/32 is subnetted, 1 subnets
C 192.168.0.4/32 is directly connected, Loopback0
      203.0.113.0/24 is variably subnetted, 2 subnets, 2 masks
C       203.0.113.0/24 is directly connected, FastEthernet1/1
L       203.0.113.1/32 is directly connected, FastEthernet1/1
S* 0.0.0.0/0 [1/0] via 203.0.113.2
```

Check the routing table on R3 next.

```
R3#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route
```

Gateway of last resort is not set

```
10.0.0.0/8 is variably subnetted, 10 subnets, 2 masks
R    10.0.0.0/24 [120/3] via 10.1.1.1, 00:00:13, FastEthernet0/0
R    10.0.1.0/24 [120/3] via 10.1.1.1, 00:00:13, FastEthernet0/0
R    10.0.2.0/24 [120/3] via 10.1.1.1, 00:00:13, FastEthernet0/0
R    10.0.3.0/24 [120/2] via 10.1.1.1, 00:00:13, FastEthernet0/0
C    10.1.0.0/24 is directly connected, FastEthernet0/1
L    10.1.0.1/32 is directly connected, FastEthernet0/1
C    10.1.1.0/24 is directly connected, FastEthernet0/0
L    10.1.1.2/32 is directly connected, FastEthernet0/0
R    10.1.2.0/24 [120/1] via 10.1.1.1, 00:00:13, FastEthernet0/0
R    10.1.3.0/24 [120/1] via 10.1.1.1, 00:00:13, FastEthernet0/0
192.168.0.0/32 is subnetted, 1 subnets
C    192.168.0.3/32 is directly connected, Loopback0
```

It is receiving routes from R4 but not from R2. Check the RIP configuration on R2.

```
R2#sh run | sec rip
R2#
```

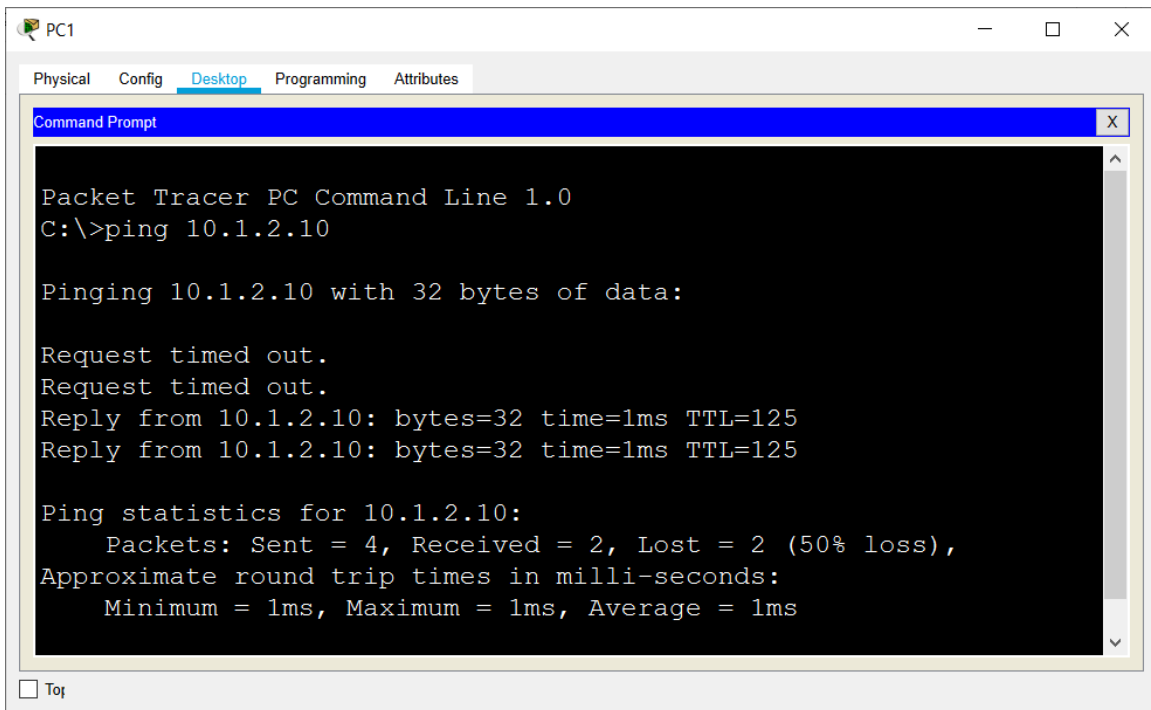
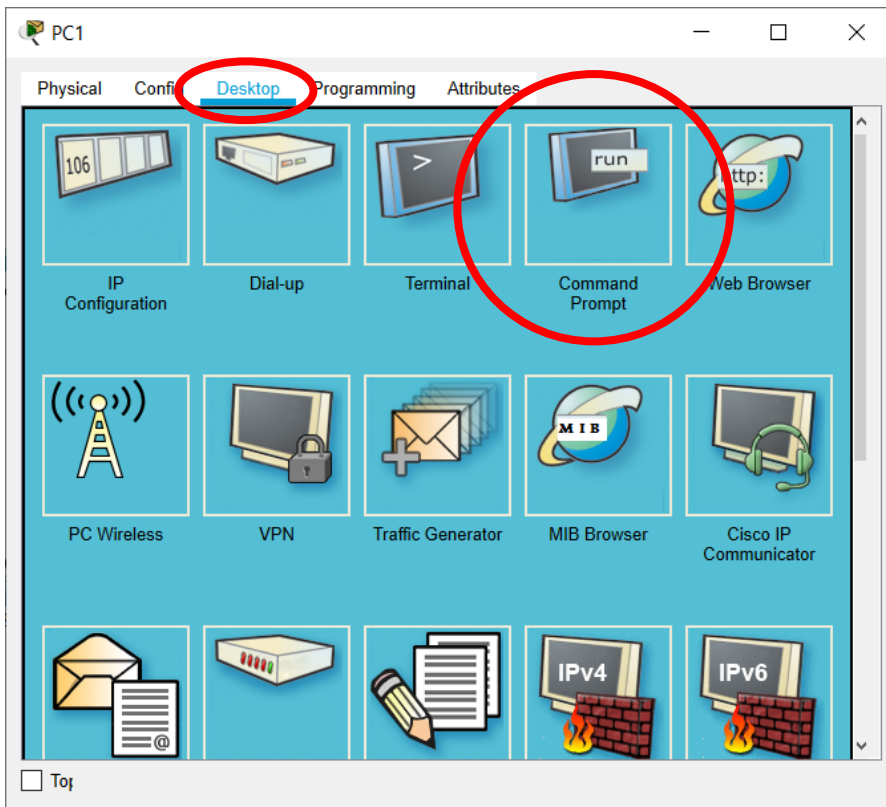
It looks like RIP is not configured for RIP. Verify this with the 'show ip protocols' command.

```
R2#sh ip protocols
Routing Protocol is "eigrp 10 "
! truncated
```

Only EIGRP is configured. Enable RIPv2.

```
R2(config)#router rip
R2(config-router)#version 2
R2(config-router)#no auto-summary
R2(config-router)#network 10.0.0.0
```

All routers should now have all 10.x.x.x routes in their routing tables. Verify this with 'show ip route' on each router and check PC1 can ping PC3.



EIGRP Troubleshooting

- 1) EIGRP has been configured on the network but the routers are not learning remote routes.

Your task is to troubleshoot the configuration and ensure that all routers have routes to every 10.x.x.x network.

There is a configuration error on each router. You will need to find and fix all errors before the network will work properly.

Note that some commands may be disabled during the CCNA exam so you should know the different troubleshooting commands.

You must not use the 'show running-configuration' command (or any derivative of it) in this EIGRP section.

There is not a set order of actions to troubleshoot this scenario. Troubleshooting in a logical fashion will however make it easier and quicker. This is how I would do it.

Check to verify that there are no EIGRP routes on any router.

```
R1#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route
```

Gateway of last resort is not set

```
10.0.0.0/8 is variably subnetted, 12 subnets, 2 masks
C    10.0.0.0/24 is directly connected, FastEthernet0/0
L    10.0.0.1/32 is directly connected, FastEthernet0/0
C    10.0.1.0/24 is directly connected, FastEthernet0/1
L    10.0.1.1/32 is directly connected, FastEthernet0/1
C    10.0.2.0/24 is directly connected, FastEthernet1/0
L    10.0.2.1/32 is directly connected, FastEthernet1/0
C    10.0.3.0/24 is directly connected, FastEthernet1/1
L    10.0.3.1/32 is directly connected, FastEthernet1/1
R    10.1.0.0/24 [120/1] via 10.0.0.2, 00:00:06, FastEthernet0/0
R    10.1.1.0/24 [120/2] via 10.0.3.2, 00:00:02, FastEthernet1/1
      [120/2] via 10.0.0.2, 00:00:06, FastEthernet0/0
R    10.1.2.0/24 [120/2] via 10.0.3.2, 00:00:02, FastEthernet1/1
R    10.1.3.0/24 [120/1] via 10.0.3.2, 00:00:02, FastEthernet1/1
192.168.0.0/32 is subnetted, 1 subnets
C    192.168.0.1/32 is directly connected, Loopback0
```

None of the routers have any EIGRP routes.

'show ip protocols' shows that EIGRP AS 100 has been enabled on all routers, except R2 which has been configured for AS 10. Correct the AS number on R2.

```
R2#show ip protocols
```

Routing Protocol is "eigrp 10"

```
R2(config)#no router eigrp 10
R2(config)#router eigrp 100
R2(config-router)#network 10.0.0.0
```

'show ip eigrp neighbors' still shows no entries after the AS number has been corrected on R2.

```
R2#sh ip eigrp neighbors
EIGRP-IPv4 Neighbors for process 100
```

On R1, 'show ip protocols' shows it has been configured with passive interfaces. It will not form adjacencies on these links.

```
R1#sh ip protocols
```

```
Routing Protocol is "eigrp 100"
```

```
Outgoing update filter list for all interfaces is not set
```

```
Incoming update filter list for all interfaces is not set
```

```
Default networks flagged in outgoing updates
```

```
Default networks accepted from incoming updates
```

```
EIGRP-IPv4 Protocol for AS(100)
```

```
Metric weight K1=1, K2=0, K3=1, K4=0, K5=0
```

```
NSF-aware route hold timer is 240
```

```
Router-ID: 10.0.3.1
```

```
Topology : 0 (base)
```

```
Active Timer: 3 min
```

```
Distance: internal 90 external 170
```

```
Maximum path: 4
```

```
Maximum hopcount 100
```

```
Maximum metric variance 1
```

```
Automatic Summarization: disabled
```

```
Maximum path: 4
```

```
Routing for Networks:
```

```
10.0.0.0
```

```
Passive Interface(s):
```

```
Passive Interface(s):
```

```
FastEthernet0/0
```

```
FastEthernet1/1
```

```
Routing Information Sources:
```

```
Gateway Distance Last Update
```

```
Distance: internal 90 external 170
```

Remove the passive interface configuration.

```
R1(config)#router eigrp 100
```

```
R1(config-router)#no passive-interface f0/0
```

```
R1(config-router)#no passive-interface f1/1
```

The adjacency between R1 and R2 should now come up.

```
R1#show ip eigrp neighbors
```

```
EIGRP-IPv4 Neighbors for AS(100)
```

H	Address	Interface	Hold (sec)	Uptime	SRTT (ms)	RT0	Q Cnt	Seq Num
0	10.0.0.2	Fa0/0	11	00:01:43	1555	5000	0	3

They will exchange routes with each other (indicated by the 'D' code on R1 for its route to 10.1.0.0/24), but the EIGRP routes from R3, R4 and R5 are still missing.

```
R1#sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route
```

Gateway of last resort is not set

```

    10.0.0.0/8 is variably subnetted, 12 subnets, 2 masks
C       10.0.0.0/24 is directly connected, FastEthernet0/0
L       10.0.0.1/32 is directly connected, FastEthernet0/0
C       10.0.1.0/24 is directly connected, FastEthernet0/1
L       10.0.1.1/32 is directly connected, FastEthernet0/1
C       10.0.2.0/24 is directly connected, FastEthernet1/0
L       10.0.2.1/32 is directly connected, FastEthernet1/0
C       10.0.3.0/24 is directly connected, FastEthernet1/1
L       10.0.3.1/32 is directly connected, FastEthernet1/1
D       10.1.0.0/24 [90/30720] via 10.0.0.2, 00:00:40, FastEthernet0/0
R       10.1.1.0/24 [120/2] via 10.0.3.2, 00:00:13, FastEthernet1/1
          [120/2] via 10.0.0.2, 00:00:20, FastEthernet0/0
R       10.1.2.0/24 [120/2] via 10.0.3.2, 00:00:13, FastEthernet1/1
R       10.1.3.0/24 [120/1] via 10.0.3.2, 00:00:13, FastEthernet1/1
    192.168.0.0/32 is subnetted, 1 subnets
C       192.168.0.1/32 is directly connected, Loopback0
```


EIGRP has been enabled on R3 for interfaces in the range 10.0.0.0/24.

```
R3#sh ip protocols
```

```
Routing Protocol is "eigrp 100"
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  Default networks flagged in outgoing updates
  Default networks accepted from incoming updates
  EIGRP-IPv4 Protocol for AS(100)
    Metric weight K1=1, K2=0, K3=1, K4=0, K5=0
    NSF-aware route hold timer is 240
    Router-ID: 10.1.1.2
    Topology : 0 (base)
      Active Timer: 3 min
      Distance: internal 90 external 170
      Maximum path: 4
      Maximum hopcount 100
      Maximum metric variance 1

  Automatic Summarization: disabled
  Maximum path: 4
Routing for Networks:
  10.0.0.0/24
  Routing Information Sources:
    Gateway          Distance      Last Update
  Distance: internal 90 external 170
```

But the interfaces have IP addresses 10.1.1.2 and 10.1.0.1. The interfaces do not fall in the range 10.0.0.0/24 so EIGRP will not be enabled.

```
R3#show ip int brief
Interface          IP-Address      OK? Method Status
Protocol
FastEthernet0/0    10.1.1.2        YES NVRAM  up
FastEthernet0/1    10.1.0.1        YES NVRAM  up
! truncated
```

Correct the network statement in the EIGRP configuration.

```
R3(config)#router eigrp 100
R3(config-router)#no network 10.0.0.0 0.0.0.255
R3(config-router)#network 10.0.0.0
```

R3 should now peer with and exchange routes with R2.

```
R3#show ip eigrp nei
```

```
EIGRP-IPv4 Neighbors for AS(100)
```

H	Address	Interface	Hold (sec)	Uptime	SRTT (ms)	RT0	Q Cnt	Seq Num
0	10.1.0.2	Fa0/1	14	00:00:40	1581	5000	0	7

```
R3#show ip route
```

```
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route
```

```
Gateway of last resort is not set
```

```
10.0.0.0/8 is variably subnetted, 10 subnets, 2 masks
D    10.0.0.0/24 [90/30720] via 10.1.0.2, 00:00:35, FastEthernet0/1
D    10.0.1.0/24 [90/33280] via 10.1.0.2, 00:00:35, FastEthernet0/1
D    10.0.2.0/24 [90/33280] via 10.1.0.2, 00:00:35, FastEthernet0/1
D    10.0.3.0/24 [90/263680] via 10.1.0.2, 00:00:35, FastEthernet0/1
C    10.1.0.0/24 is directly connected, FastEthernet0/1
L    10.1.0.1/32 is directly connected, FastEthernet0/1
C    10.1.1.0/24 is directly connected, FastEthernet0/0
L    10.1.1.2/32 is directly connected, FastEthernet0/0
R    10.1.2.0/24 [120/1] via 10.1.1.1, 00:00:11, FastEthernet0/0
R    10.1.3.0/24 [120/1] via 10.1.1.1, 00:00:11, FastEthernet0/0
192.168.0.0/32 is subnetted, 1 subnets
C    192.168.0.3/32 is directly connected, Loopback0
```

'Show ip protocols' on R4 shows that EIGRP AS 100 has been enabled but it is not routing for any networks.

```
R4#sh ip proto
```

```
Routing Protocol is "eigrp 100"
```

```
Outgoing update filter list for all interfaces is not set
```

```
Incoming update filter list for all interfaces is not set
```

```
Default networks flagged in outgoing updates
```

```
Default networks accepted from incoming updates
```

```
EIGRP-IPv4 Protocol for AS(100)
```

```
Metric weight K1=1, K2=0, K3=1, K4=0, K5=0
```

```
NSF-aware route hold timer is 240
```

```
Router-ID: 203.0.113.1
```

```
Topology : 0 (base)
```

```
Active Timer: 3 min
```

```
Distance: internal 90 external 170
```

```
Maximum path: 4
```

```
Maximum hopcount 100
```

```
Maximum metric variance 1
```

```
Automatic Summarization: disabled
```

```
Maximum path: 4
```

Routing for Networks:

```
Routing Information Sources:
```

```
Gateway Distance Last Update
```

```
Distance: internal 90 external 170
```

Add the network statement to enable EIGRP on the interfaces.

```
R4(config)#router eigrp 100
```

```
R4(config-router)# network 10.0.0.0
```

R4 should now peer with and exchange routes with R3.

```
R4#sh ip eigrp nei
```

```
EIGRP-IPv4 Neighbors for AS(100)
```

H	Address	Interface	Hold (sec)	Uptime	SRTT (ms)	RT0	Q Cnt	Seq Num
0	10.1.1.2	Fa0/0	11	00:00:27	23	138	0	7

R4#show ip route

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
* - candidate default, U - per-user static route, o - ODR
P - periodic downloaded static route

Gateway of last resort is 203.0.113.2 to network 0.0.0.0

10.0.0.0/8 is variably subnetted, 11 subnets, 2 masks
D 10.0.0.0/24 [90/33280] via 10.1.1.2, 00:00:24, FastEthernet0/0
D 10.0.1.0/24 [90/35840] via 10.1.1.2, 00:00:24, FastEthernet0/0
D 10.0.2.0/24 [90/35840] via 10.1.1.2, 00:00:24, FastEthernet0/0
D 10.0.3.0/24 [90/266240] via 10.1.1.2, 00:00:24, FastEthernet0/0
D 10.1.0.0/24 [90/30720] via 10.1.1.2, 00:00:24, FastEthernet0/0
C 10.1.1.0/24 is directly connected, FastEthernet0/0
L 10.1.1.1/32 is directly connected, FastEthernet0/0
C 10.1.2.0/24 is directly connected, FastEthernet0/1
L 10.1.2.1/32 is directly connected, FastEthernet0/1
C 10.1.3.0/24 is directly connected, FastEthernet1/0
L 10.1.3.1/32 is directly connected, FastEthernet1/0
192.168.0.0/32 is subnetted, 1 subnets
C 192.168.0.4/32 is directly connected, Loopback0
203.0.113.0/24 is variably subnetted, 2 subnets, 2 masks
C 203.0.113.0/24 is directly connected, FastEthernet1/1
L 203.0.113.1/32 is directly connected, FastEthernet1/1
S* 0.0.0.0/0 [1/0] via 203.0.113.2

EIGRP has been enabled on R5 for interfaces in the range 100.0.0.0/24.

```
R5#sh ip protocols
```

```
Routing Protocol is "eigrp 100"
```

```
Outgoing update filter list for all interfaces is not set
```

```
Incoming update filter list for all interfaces is not set
```

```
Default networks flagged in outgoing updates
```

```
Default networks accepted from incoming updates
```

```
EIGRP-IPv4 Protocol for AS(100)
```

```
Metric weight K1=1, K2=0, K3=1, K4=0, K5=0
```

```
NSF-aware route hold timer is 240
```

```
Router-ID: 10.1.3.2
```

```
Topology : 0 (base)
```

```
Active Timer: 3 min
```

```
Distance: internal 90 external 170
```

```
Maximum path: 4
```

```
Maximum hopcount 100
```

```
Maximum metric variance 1
```

```
Automatic Summarization: disabled
```

```
Maximum path: 4
```

```
Routing for Networks:
```

```
100.0.0.0
```

```
Routing Information Sources:
```

```
Gateway Distance Last Update
```

```
Distance: internal 90 external 170
```

This is a typo and should have been 10.0.0.0.

```
R5#show ip interface brief
```

Interface	IP-Address	OK?	Method	Status	Protocol
FastEthernet0/0	10.1.3.2	YES	manual	up	up
FastEthernet0/1	10.0.3.2	YES	manual	up	up
FastEthernet1/0	unassigned	YES	manual	administratively down	down
FastEthernet1/1	unassigned	YES	NVRAM	administratively down	down
Loopback0	192.168.0.5	YES	manual	up	up
Vlan1	unassigned	YES	unset	administratively down	down

Correct the network statement in the EIGRP configuration.

```
R5(config)#router eigrp 100
```

```
R5(config-router)#no network 100.0.0.0
```

```
R5(config-router)#network 10.0.0.0
```

R5 should now peer with and exchange routes with R1 and R4.

```
R5#sh ip eigrp nei
```

```
EIGRP-IPv4 Neighbors for AS(100)
```

H	Address	Interface	Hold (sec)	Uptime	SRTT (ms)	RT0	Q Cnt	Seq Num
1	10.0.3.1	Fa0/1	13	00:00:25	44	264	0	10
0	10.1.3.1	Fa0/0	12	00:00:25	1112	5000	0	8

```
R5#show ip route
```

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
* - candidate default, U - per-user static route, o - ODR
P - periodic downloaded static route

Gateway of last resort is not set

```
10.0.0.0/8 is variably subnetted, 10 subnets, 2 masks
D    10.0.0.0/24 [90/30720] via 10.0.3.1, 00:00:39, FastEthernet0/1
D    10.0.1.0/24 [90/30720] via 10.0.3.1, 00:00:39, FastEthernet0/1
D    10.0.2.0/24 [90/30720] via 10.0.3.1, 00:00:39, FastEthernet0/1
C    10.0.3.0/24 is directly connected, FastEthernet0/1
L    10.0.3.2/32 is directly connected, FastEthernet0/1
D    10.1.0.0/24 [90/33280] via 10.0.3.1, 00:00:39, FastEthernet0/1
D    10.1.1.0/24 [90/35840] via 10.0.3.1, 00:00:39, FastEthernet0/1
D    10.1.2.0/24 [90/38400] via 10.0.3.1, 00:00:39, FastEthernet0/1
C    10.1.3.0/24 is directly connected, FastEthernet0/0
L    10.1.3.2/32 is directly connected, FastEthernet0/0
192.168.0.0/32 is subnetted, 1 subnets
C    192.168.0.5/32 is directly connected, Loopback0
```

EIGRP routes should now have replaced the RIP routes on all routers.
Verify this with 'show ip route' on each router and check PC1 can still ping PC3.

```
R1#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route
```

Gateway of last resort is not set

```

    10.0.0.0/8 is variably subnetted, 12 subnets, 2 masks
C       10.0.0.0/24 is directly connected, FastEthernet0/0
L       10.0.0.1/32 is directly connected, FastEthernet0/0
C       10.0.1.0/24 is directly connected, FastEthernet0/1
L       10.0.1.1/32 is directly connected, FastEthernet0/1
C       10.0.2.0/24 is directly connected, FastEthernet1/0
L       10.0.2.1/32 is directly connected, FastEthernet1/0
C       10.0.3.0/24 is directly connected, FastEthernet1/1
L       10.0.3.1/32 is directly connected, FastEthernet1/1
D       10.1.0.0/24 [90/30720] via 10.0.0.2, 00:09:47, FastEthernet0/0
D       10.1.1.0/24 [90/33280] via 10.0.0.2, 00:06:23, FastEthernet0/0
D       10.1.2.0/24 [90/35840] via 10.0.0.2, 00:04:08, FastEthernet0/0
D       10.1.3.0/24 [90/261120] via 10.0.3.2, 00:01:39, FastEthernet1/1
    192.168.0.0/32 is subnetted, 1 subnets
C       192.168.0.1/32 is directly connected, Loopback0
```

